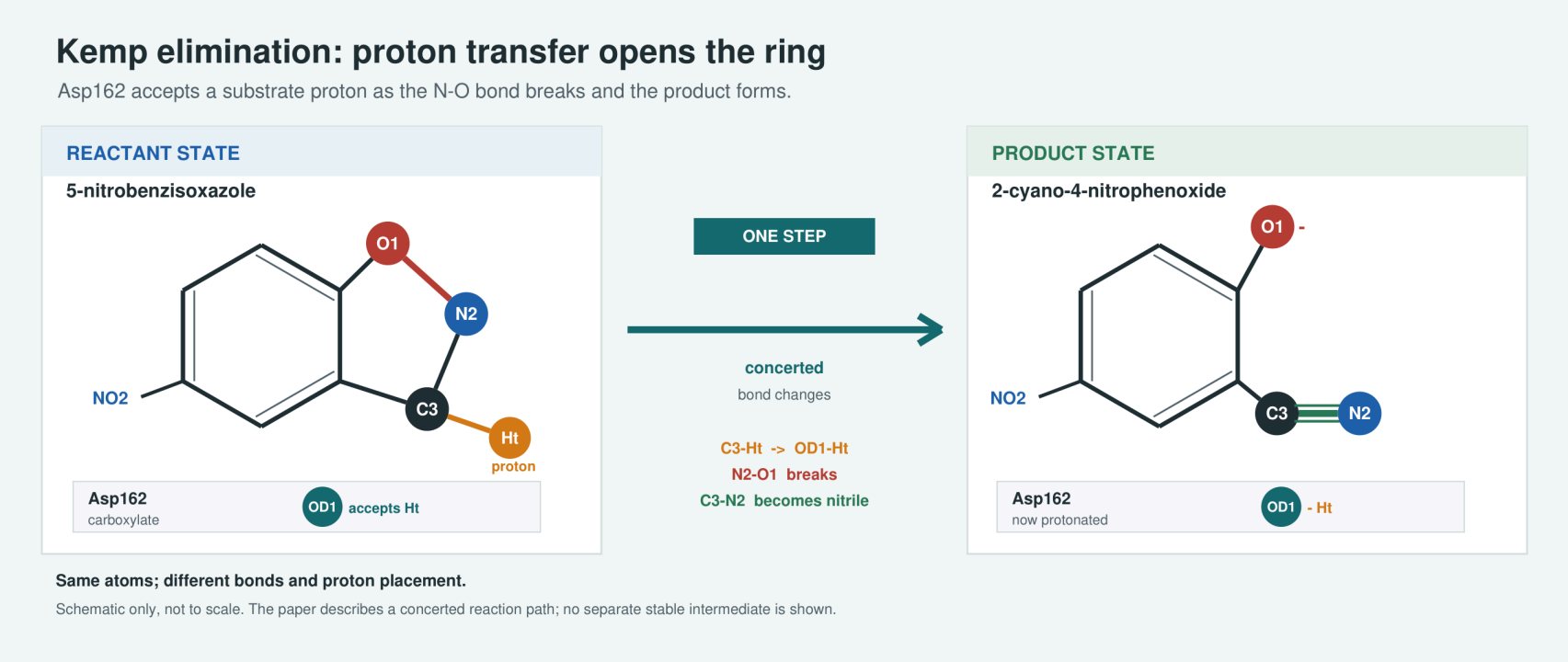
REACTION AT A GLANCE

Kemp elimination: what changes?

Asp162 accepts a proton from 5-nitrobenzisoxazole as the substrate ring opens to form the cyanophenoxide product.



Atom labels: C3 is the substrate carbon that loses Ht; Ht is the transferred hydrogen; O1 and N2 are the ring oxygen and nitrogen. Asp162 is enzyme aspartate residue 162; OD1 is one of its side-chain oxygens and accepts Ht.

UNDERSTANDING EVB

Two bonding descriptions, one reaction

REACTANT STATE (R)

Before

The substrate still has its C-H bond and closed N-O-containing ring. Asp162 is a carboxylate ready to accept the proton.

PRODUCT STATE (P)

After

Asp162 has accepted the proton. The N-O bond is broken, the nitrile has formed, and the ring-opened phenoxide product remains.

EVB

How the states mix

EVB combines energy descriptions for R and P. The energy offset shifts one state relative to the other; coupling controls how strongly they mix.

Explore the saved examples in this order

1	**EVB fundamentals**	Change coupling and offset one at a time.
2	**Saved reaction profile**	Find the reactant, high-energy region, and product.
3	**Parameter comparison**	Compare parameter pairs and their product energies.

Paper Figure 3e shows barrier estimates; the OpenMM viewer shows a free-energy profile. Different boundaries, electrostatics, Morse references, and sampling mean these are not a direct comparison.

FULL CALCULATION WORKFLOW

From a structure to a free-energy profile

1 · PREPARE

Build matched R and P

Use the same atoms in the same order. Assign state-specific bonds, charges, force-field terms, and EVB settings. Check the atom map, total charge, proton position, solvent, and ions before running.

2 · RELAX

Stabilize both endpoints

Minimize and equilibrate R and P separately. Inspect geometry, contacts, and energy stability. Relaxation prepares each endpoint; it does not calculate the reaction barrier.

3 · FEP/PROFILE

Sample the lambda path

Sample a set of lambda windows between the state descriptions. Check neighboring-window overlap and convergence, compare independent replicas, then analyze the barrier and reaction free energy.

What this workshop contains

This repository includes saved Kemp data and browser exercises for learning the method. It does not include the complete live R/P builder, endpoint-relaxation inputs, or lambda-window run scripts, so the three simulation stages above are explained but not launched here.